

YEAR 8 ADVANCED → YEAR 9 ADVANCED

Summer Maths

Ready for Year 9

Six weeks. Twenty questions a week. Every Year 8 skill that Year 9 Advanced builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always fractions and integers, question 11 is always Pythagoras, and so on. If the same number keeps going wrong, that is the skill to revise before Year 9.

1	Fractions & integers	11	Pythagoras
2	Percentages & rounding	12	Area
3	Indices	13	Volume
4	Roots & surds	14	Number plane
5	Simplify & substitute	15	Gradient
6	Expand	16	y = mx + c
7	Factorise	17	Angles
8	Equations	18	Probability
9	Equations	19	Algebra in context
10	Formulas	20	Measurement in context

Why Julia? Each banner is a *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Change c and you get a completely different shape – the same rule as the Mandelbrot set, seen from a different angle.

Rules of the game

- **No calculator.** Leave answers exact (e.g. $\sqrt{34}$ or 9π) unless asked to round.
- **Show working** in the box under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 9 (Stage 5) work. Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 9 Advanced: products and factors (expanding and factorising), surds and Pythagoras, indices, equations, trigonometry and coordinate geometry. Every one of them starts from the skills in these six pages.

WEEK 1 — Summer Maths
Ready for Year 9

Date started _____
Score ____ / 20
Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS
Evaluate $\frac{3}{4} + \frac{2}{3}$

2 PERCENTAGES & ROUNDING
Increase $\$80$ by 15%.

3 INDICES
Write as a single power: $2^5 \times 2^3 \div 2^4$

4 ROOTS & SURDS
Evaluate $\sqrt{36} + \sqrt[3]{8}$

5 SIMPLIFY & SUBSTITUTE
Simplify $7a + 3b - 2a + b$

6 EXPAND
Expand $4(3x - 5)$

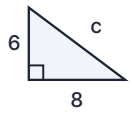
7 FACTORISE
Factorise $6x + 9$

8 EQUATIONS
Solve $3x - 7 = 11$

9 EQUATIONS
Solve $5x + 2 = 2x + 17$

10 FORMULAS
$P = 2(l + w)$. Find P when $l = 8$ and $w = 3.5$.

11 PYTHAGORAS
Find c. (cm)



12 AREA
Find the area of a trapezium with parallel sides 6 cm and 10 cm and height 4 cm.

13 VOLUME
Find the volume of a rectangular prism 6 cm by 5 cm by 4 cm.

14 NUMBER PLANE
Complete the table for $y = 2x - 1$.

x	-1	0	1	2
y				

15 GRADIENT ★
Find the gradient of the interval joining $(1, 2)$ and $(3, 8)$.

16 Y = MX + C ★
State the gradient and y-intercept of $y = 3x - 4$.

17 ANGLES
Find a.



18 PROBABILITY
A fair die is rolled. Find $P(\text{number greater than } 4)$.

19 ALGEBRA IN CONTEXT
Mia has $\$40$ and saves $\$15$ each week of the holidays. Write a rule for her savings S after w weeks, then find S after 6 weeks.

20 MEASUREMENT IN CONTEXT
A rectangular pool area, 12 m by 7.5 m, is fenced on all four sides. How much fencing is needed?

BONUS Find two consecutive whole numbers whose squares differ by 15.

WEEK 2

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS

Evaluate $-8 + (-5) \times 3$

2 PERCENTAGES & ROUNDING

Round 12.6849 to 2 decimal places.

3 INDICES

Write 72 as a product of prime factors in index form.

4 ROOTS & SURDS ★

Simplify $\sqrt{50}$

5 SIMPLIFY & SUBSTITUTE

If $a = -2$, $b = 5$: $a^2 - 3b$

6 EXPAND

Expand and simplify $3(x + 4) + 2(x - 1)$

7 FACTORISE

Factorise $8a^2 - 12a$

8 EQUATIONS

Solve $\frac{x}{4} + 3 = 8$

9 EQUATIONS

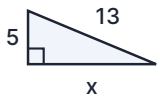
Solve $3(x - 2) = 15$

10 FORMULAS

$v = u + at$. Find a when $v = 30$, $u = 6$ and $t = 8$.

11 PYTHAGORAS

Find x. (cm)



12 AREA

Find the area of a circle with radius 5 cm, in terms of π.

13 VOLUME

Find the volume of a cylinder with radius 2 cm and height 10 cm, in terms of π.

14 NUMBER PLANE

Does the point $(3, 7)$ lie on the line $y = 3x - 2$?

15 GRADIENT ★

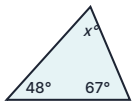
Find the gradient of the interval joining $(0, 5)$ and $(4, -3)$.

16 Y = MX + C ★

Write the equation of the line with gradient 2 and y-intercept -5.

17 ANGLES

Find x.



18 PROBABILITY

A bag has 3 red, 5 blue and 2 green marbles. Find $P(\text{not blue})$.

19 ALGEBRA IN CONTEXT

Ella pays $\$x$ towards a beach umbrella, Jack pays $\$5$ more than Ella and Sam pays twice what Ella pays. The umbrella costs $\$45$. Find x.

20 MEASUREMENT IN CONTEXT

A triangular sail has base 2.4 m and perpendicular height 3 m. Find its area.

BONUS Solve $x^2 = 2x$. (Careful – there are two answers.)

WEEK 3 | Summer Maths
Ready for Year 9

Date started _____
Score ____ / 20
Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS
Evaluate $2\frac{1}{2} \times 1\frac{3}{5}$.

2 PERCENTAGES & ROUNDING
Write $\frac{3}{8}$ as a percentage.

3 INDICES
Evaluate $(3^2)^3 \div 3^4$

4 ROOTS & SURDS
Evaluate $\sqrt{3} \times \sqrt{12}$

5 SIMPLIFY & SUBSTITUTE
Simplify $12x^2y \div 4xy$

6 EXPAND ★
Expand $(x+3)(x+5)$

7 FACTORISE
Factorise $15xy - 10x$

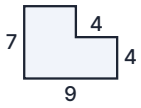
8 EQUATIONS
Solve $7 - 2x = 1$

9 EQUATIONS
Solve $6x - 4 = 2x + 10$

10 FORMULAS
$A = \frac{1}{2}bh$. Find b when $A = 24$ and $h = 6$.

11 PYTHAGORAS
Is a triangle with sides 7, 24 and 25 right-angled?

12 AREA
Find the area. (cm)



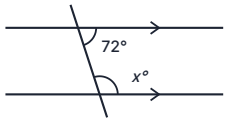
13 VOLUME
A triangular prism has a triangle of base 6 cm and height 4 cm as its cross-section, and length 12 cm. Find its volume.

14 NUMBER PLANE
For $y = -2x + 5$, find y when $x = 3$.

15 GRADIENT ★
Find the gradient of the interval joining $(-2, 1)$ and $(4, 4)$.

16 Y = MX + C ★
Which line is steeper: $y = 2x + 7$ or $y = 5x - 1$? Why?

17 ANGLES
The lines are parallel. Find x.



18 PROBABILITY
The probability that a spinner lands on red is 0.35. Find $P(\text{not red})$.

19 ALGEBRA IN CONTEXT
Kayak hire on the Tweed River costs $20 plus $15 per hour. Lily paid $95. Write an equation and find how many hours she hired it for.

20 MEASUREMENT IN CONTEXT
A sandpit 3 m by 2 m is filled with sand 0.25 m deep. Find the volume of sand.

BONUS Show that the sum of any three consecutive whole numbers is always divisible by 3.

WEEK 4 — Summer Maths
Ready for Year 9

Date started _____
Score ____ / 20
Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS
Evaluate $(-3)^2 - 4 \times (-2)$

2 PERCENTAGES & ROUNDING
Decrease \$250 by 20%.

3 INDICES
Evaluate $5^0 + 2^3 \times 3$

4 ROOTS & SURDS
Estimate $\sqrt{70}$ to 1 decimal place. (Check by squaring.)

5 SIMPLIFY & SUBSTITUTE
Write an expression for "5 less than three times n".

6 EXPAND
Expand and simplify $x(2x + 7) - 3x$

7 FACTORISE
Factorise $9p + 12q$

8 EQUATIONS
Solve $\frac{2x}{5} = 6$

9 EQUATIONS
Solve $3x + 8 = 5x - 4$

10 FORMULAS ★
Make x the subject of $y = 4x - 3$.

11 PYTHAGORAS
A 2.5 m ladder leans against a wall with its foot 0.7 m from the wall. How high does it reach?

12 AREA
Find the area of a kite with diagonals 8 cm and 11 cm.

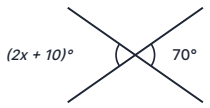
13 VOLUME
Find the volume of a cylinder with diameter 6 cm and height 5 cm, in terms of π.

14 NUMBER PLANE
Which point is **not** on $y = x + 4$: $(0, 4)$, $(2, 6)$, $(-1, 5)$, $(-4, 0)$?

15 GRADIENT ★
Find the gradient of the interval joining $(-3, 5)$ and $(6, 5)$. What kind of line is it?

16 Y = MX + C ★
Write $2x + y = 7$ in the form $y = mx + c$ and state the gradient.

17 ANGLES
Find x.



18 PROBABILITY
A letter is chosen at random from the word KINGSCLIFF. Find $P(\text{F})$.

19 ALGEBRA IN CONTEXT
A triangle has sides x, $x + 3$ and $2x - 1$ cm. Its perimeter is 30 cm. Find x.

20 MEASUREMENT IN CONTEXT
A circular garden bed has diameter 6 m. Find its area in terms of π.

BONUS A square has area $50\ \text{cm}^2$. Find its side length in simplest surd form, and the length of its diagonal.

WEEK 5 | Summer Maths
Ready for Year 9

Date started _____
Score ____ / 20
Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS
Evaluate $\frac{5}{6} - \frac{1}{4}$

2 PERCENTAGES & ROUNDING ★
Round 0.04682 to 2 significant figures.

3 INDICES
Evaluate $\frac{7^5}{7^3}$

4 ROOTS & SURDS
Evaluate $\sqrt{16 \times 25}$

5 SIMPLIFY & SUBSTITUTE
If $p = 3,\ q = -4$: $(p - q)^2$

6 EXPAND
Expand and simplify $5(2a - 1) - 3a$

7 FACTORISE
Factorise $12xy + 18y^2$

8 EQUATIONS
Solve $\frac{x}{3} - 4 = 2$

9 EQUATIONS
Solve $2(x + 5) = 18$

10 FORMULAS
$C = 2\pi r$. Find r when $C = 18\pi$.

11 PYTHAGORAS
Find the exact length of the hypotenuse if the shorter sides are 3 and 5.

12 AREA
Find the area of a semicircle with diameter 8 cm, in terms of π.

13 VOLUME
A prism has volume $96\ \text{cm}^3$ and length 8 cm. Find the area of its cross-section.

14 NUMBER PLANE
Find the point where $y = x + 1$ and $y = 7 - x$ meet.

15 GRADIENT ★
Find the gradient of the interval joining $(2,\ -1)$ and $(5,\ 11)$.

16 Y = MX + C ★
Write the equation of the line with gradient -4 that passes through $(0,\ 3)$.

17 ANGLES
Three angles of a quadrilateral are 95°, 80° and 110°. Find the fourth.

18 PROBABILITY
In a class of 28, 12 students have been to a Gold Coast theme park. A student is chosen at random. Find $P(\text{has not been})$.

19 ALGEBRA IN CONTEXT
Ava is 4 years older than her brother. In 3 years their ages will add to 30. How old is Ava now?

20 MEASUREMENT IN CONTEXT
An esky measures 50 cm by 30 cm by 30 cm inside. Find its capacity in litres. ($1000\ \text{cm}^3 = 1$ L)

BONUS Expand and simplify $(x + 2)^2 - (x - 2)^2$

WEEK 6 — Summer Maths
Ready for Year 9

Date started _____
Score ____ / 20
Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Linear · Angles & chance · In context · ★ Year 9 preview (Stage 5) · No calculator

1 FRACTIONS & INTEGERS
Evaluate $-24 \div (-6) + 3 \times (-2)$

2 PERCENTAGES & ROUNDING
A \$120 cricket bat is on sale for \$90. What is the percentage discount?

3 INDICES
Evaluate $2^4 \times 5^4$

4 ROOTS & SURDS
Between which two consecutive whole numbers does $\sqrt{90}$ lie?

5 SIMPLIFY & SUBSTITUTE
Simplify $4ab \times 3a \div 6b$

6 EXPAND ★
Expand $(2x - 1)(x + 4)$

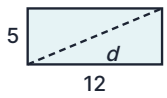
7 FACTORISE
Factorise $9m^2 + 6m$

8 EQUATIONS
Solve $5 - 3x = 17$

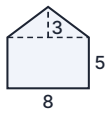
9 EQUATIONS
Solve $\dfrac{x + 4}{3} = 5$

10 FORMULAS
$E = \frac{1}{2}mv^2$. Find E when $m = 4$ and $v = 6$.

11 PYTHAGORAS
Find the length of the diagonal d. (cm)



12 AREA
Find the area. (cm)



13 VOLUME
A prism has a trapezium cross-section (parallel sides 3 cm and 7 cm, height 4 cm) and length 10 cm. Find its volume.

14 NUMBER PLANE
Find the rule linking x and y.

x	0	1	2	3
y	5	8	11	14

15 GRADIENT ★
Find the gradient of the interval joining $(-4, 7)$ and $(2, -2)$.

16 Y = MX + C ★
A line passes through $(0, -2)$ and $(3, 4)$. Find its equation.

17 ANGLES
An isosceles triangle has an apex angle of 40°. Find each base angle.

18 PROBABILITY
A whole number from 1 to 20 is chosen at random. Find $P(\text{multiple of 3 or 5})$.

19 ALGEBRA IN CONTEXT
A cricket team scored x runs in its first innings and 45 fewer in its second. The total was 215. Find x.

20 MEASUREMENT IN CONTEXT
A circular trampoline has radius 2 m. A safety net goes around its edge. Find the length of the net in terms of π.

BONUS A right-angled triangle has sides x, $x + 7$ and $x + 8$. Find x.

Hints & formulas

Number

$\frac{a}{b} + \frac{c}{d}$: use a common denominator

$\times$: multiply tops and bottoms; $\div$: flip the second, then multiply

Increase by 15%: $\times 1.15$ Decrease by 20%: $\times 0.8$

$$\%\text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$$

★ Significant figures start at the first non-zero digit

Indices & roots

$a^m \times a^n = a^{m+n}$ $a^m \div a^n = a^{m-n}$

$(a^m)^n = a^{mn}$ $(ab)^n = a^n b^n$ $a^0 = 1$

$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

★ $\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$

Algebra

Like terms: same letters, same powers

$a(b + c) = ab + ac$, $-(x - 5) = -x + 5$

Factorise: take out the HCF, then check by expanding

★ $(a + b)(c + d) = ac + ad + bc + bd$

Equations & formulas

Do the same to both sides

Unknowns on both sides: collect them on one side first

$x^2 = c \Rightarrow x = \pm\sqrt{c}$

Formulas: substitute, then solve

★ Change the subject: undo operations in reverse order

Measurement

Pythagoras: $c^2 = a^2 + b^2$ (c = hypotenuse)

$A = \frac{1}{2}bh \cdot A = bh \cdot A = \frac{h}{2}(a + b) \cdot A = \frac{1}{2}xy$

Circle: $C = 2\pi r = \pi d$, $A = \pi r^2$

Prism: $V = Ah$ Cylinder: $V = \pi r^2 h$

$1\text{ cm}^3 = 1\text{ mL}$, $1000\text{ cm}^3 = 1\text{ L}$

Linear relationships

A point lies on a line if its coordinates satisfy the equation

★ Gradient $m = \dfrac{\text{rise}}{\text{run}} = \dfrac{y_2 - y_1}{x_2 - x_1}$

★ $y = mx + c$: m = gradient, c = y-intercept

Horizontal lines have gradient 0; uphill (left to right) is positive

Angles & probability

Straight line 180° · point 360° · vertically opposite angles are equal

Parallel lines: corresponding and alternate angles are equal; co-interior add to 180°

Triangle 180° · quadrilateral 360°

$$P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}},\quad P(\text{not } E) = 1 - P(E)$$

Worded problems

1. Choose a letter for the unknown and say what it means.
2. Write an equation from the words.
3. Solve it.
4. Answer the question in a sentence, with units.

Answers

1 Week 1 answers

1 $\frac{17}{12}$ $\frac{9}{12}+\frac{8}{12}=\frac{17}{12}=1\frac{5}{12}$

2 \$92 15% of $80 = 12$; $80 + 12 = 92$

3 2^4 $2^{5+3-4} = 2^4\ (= 16)$

4 8 $6 + 2 = 8$

5 $5a + 4b$ $7a - 2a = 5a$; $3b + b = 4b$

6 $12x - 20$ $4 \times 3x - 4 \times 5$

7 $3(2x + 3)$ HCF $= 3$

8 $x = 6$ $3x = 18$, $x = 6$

9 $x = 5$ $3x + 2 = 17$, $3x = 15$

10 $P = 23$ $2 \times 11.5 = 23$

11 **10 cm** $c^2 = 36 + 64 = 100$

12 32 cm^2 $\frac{4}{2}(6 + 10) = 2 \times 16$

13 120 cm^3 $6 \times 5 \times 4$

14 $-3,\ -1,\ 1,\ 3$ substitute each x: e.g. $2(-1) - 1 = -3$

15 3 $\frac{\text{rise}}{\text{run}} = \frac{8-2}{3-1} = \frac{6}{2}$

16 $m = 3, c = -4$ compare with $y = mx + c$

17 $a = 55$ angles on a straight line: $180 - 125$

18 $\frac{1}{3}$ $\{5, 6\}$: $\frac{2}{6}$

19 $S = 40 + 15w$; \$130 $40 + 15 \times 6 = 130$

20 39 m $2(12 + 7.5) = 39$

B **7 and 8** $(n + 1)^2 - n^2 = 2n + 1 = 15$, so $n = 7$; check $64 - 49 = 15$

2 Week 2 answers

1 -23 $-8 + (-15) = -23$ (multiply first)

2 12.68 third decimal is 4, so round down

3 $2^3 \times 3^2$ $72 = 8 \times 9$

4 $5\sqrt{2}$ $\sqrt{25 \times 2} = 5\sqrt{2}$

5 -11 $(-2)^2 - 15 = 4 - 15$

6 $5x + 10$ $3x + 12 + 2x - 2$

7 $4a(2a - 3)$ HCF $= 4a$

8 $x = 20$ $\frac{x}{4} = 5$, $x = 20$

9 $x = 7$ $x - 2 = 5$

10 $a = 3$ $30 = 6 + 8a$, $8a = 24$

11 **12 cm** $x^2 = 169 - 25 = 144$

12 25π cm^2 $\pi \times 5^2$

13 40π cm^3 $\pi \times 2^2 \times 10$

14 **Yes** $3(3) - 2 = 7$ ✓

15 -2 $\frac{-3-5}{4-0} = \frac{-8}{4}$

16 $y = 2x - 5$ $y = mx + c$ with $m = 2$, $c = -5$

17 $x = 65$ angle sum: $180 - 48 - 67$

18 $\frac{1}{2}$ $\frac{5}{10}$ are not blue

19 $x = 10$ $x + (x + 5) + 2x = 45$, $4x = 40$

20 3.6 m^2 $\frac{1}{2} \times 2.4 \times 3$

B $x = 0$ **or** $x = 2$ $x^2 - 2x = 0$, $x(x - 2) = 0$; dividing by x loses $x = 0$

Answers

3 Week 3 answers

1. 4 $\frac{5}{2} \times \frac{8}{5} = \frac{40}{10}$
2. 37.5% $\frac{3}{8} = 0.375$
3. 9 $3^6 \div 3^4 = 3^2$
4. 6 $\sqrt{36} = 6$
5. $3x$ $\frac{12x^2y}{4xy}$
6. $x^2 + 8x + 15$ $x^2 + 5x + 3x + 15$
7. $5x(3y - 2)$ HCF $= 5x$
8. $x = 3$ $-2x = -6$
9. $x = \frac{7}{2}$ $4x = 14$, $x = 3.5$
10. $b = 8$ $24 = 3b$
11. **Yes** $49 + 576 = 625 = 25^2$
12. 51 cm^2 $9 \times 7 - 4 \times 3 = 63 - 12$
13. 144 cm^3 $A = 12$; $V = 12 \times 12$
14. $y = -1$ $-6 + 5$
15. $\frac{1}{2}$ $\frac{4 - 1}{4 - (-2)} = \frac{3}{6}$
16. $y = 5x - 1$ larger gradient: $5 > 2$
17. $x = 108$ co-interior angles: $180 - 72$
18. 0.65 $1 - 0.35$
19. **5 hours** $20 + 15h = 95$, $15h = 75$
20. 1.5 m^3 $3 \times 2 \times 0.25$

B $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$ a multiple of 3 for every n

4 Week 4 answers

1. 17 $9 + 8$
2. \$200 20% of $250 = 50$
3. 25 $1 + 8 \times 3 = 1 + 24$
4. 8.4 $8.3^2 = 68.89$, $8.4^2 = 70.56$; 70 is closer to 70.56
5. $3n - 5$ three times n is $3n$; then subtract 5
6. $2x^2 + 4x$ $2x^2 + 7x - 3x$
7. $3(3p + 4q)$ HCF $= 3$
8. $x = 15$ $2x = 30$
9. $x = 6$ $8 = 2x - 4$, $2x = 12$
10. $x = \frac{y + 3}{4}$ $y + 3 = 4x$
11. **2.4 m** $h^2 = 6.25 - 0.49 = 5.76$
12. 44 cm^2 $\frac{1}{2} \times 8 \times 11$
13. 45π cm^3 $r = 3$: $\pi \times 9 \times 5$
14. $(-1, 5)$ $-1 + 4 = 3 = 5$
15. **0; horizontal** rise $= 0$
16. $y = -2x + 7$; $m = -2$ subtract $2x$ from both sides
17. $x = 30$ vertically opposite: $2x + 10 = 70$
18. $\frac{1}{5}$ 2 F's out of 10 letters
19. $x = 7$ $4x + 2 = 30$
20. 9π m^2 $r = 3$: $\pi \times 3^2$

B $5\sqrt{2}$ **cm; 10 cm** $\sqrt{50} = 5\sqrt{2}$; diagonal$^2 = 50 + 50 = 100$

Answers

5 Week 5 answers

1 $\frac{7}{12}$ $\frac{10}{12} - \frac{3}{12}$

2 0.047 first significant figure is 4; next digit 8 rounds up

3 49 $7^{5-3} = 7^2$

4 20 4×5

5 49 $(3-(-4))^2 = 7^2$

6 $7a - 5$ $10a - 5 - 3a$

7 $6y(2x + 3y)$ HCF $= 6y$

8 $x = 18$ $\frac{x}{3} = 6$

9 $x = 4$ $x + 5 = 9$

10 $r = 9$ $2\pi r = 18\pi$

11 $\sqrt{34}$ $9 + 25 = 34$

12 8π cm^2 $\frac{1}{2} \times \pi \times 4^2$

13 12 cm^2 $96 \div 8$

14 $(3, 4)$ $x + 1 = 7 - x$, $x = 3$

15 4 $\frac{11-(-1)}{5-2} = \frac{12}{3}$

16 $y = -4x + 3$ $(0, 3)$ is the y-intercept

17 75° $360 - 285$

18 $\frac{4}{7}$ $\frac{16}{28}$

19 14 $b + 3 + b + 7 = 30$, $b = 10$

20 45 L $45\,000$ cm^3

B $8x$ $(x^2 + 4x + 4) - (x^2 - 4x + 4)$

6 Week 6 answers

1 -2 $4 + (-6)$

2 25% $\frac{30}{120} = \frac{1}{4}$

3 10000 $(2 \times 5)^4 = 10^4 = 10\,000$

4 9 **and** 10 $81 < 90 < 100$

5 $2a^2$ $\frac{12a^2b}{6b}$

6 $2x^2 + 7x - 4$ $2x^2 + 8x - x - 4$

7 $3m(3m + 2)$ HCF $= 3m$

8 $x = -4$ $-3x = 12$

9 $x = 11$ $x + 4 = 15$

10 $E = 72$ $\frac{1}{2} \times 4 \times 36$

11 13 **cm** $d^2 = 144 + 25 = 169$

12 52 cm^2 $8 \times 5 + \frac{1}{2} \times 8 \times 3$

13 200 cm^3 $A = \frac{4}{2}(3 + 7) = 20$

14 $y = 3x + 5$ y goes up by 3; starts at 5

15 $-\frac{3}{2}$ $\frac{-2-7}{2-(-4)} = \frac{-9}{6}$

16 $y = 2x - 2$ $m = \frac{6}{3} = 2$, $c = -2$

17 70° $(180 - 40) \div 2$

18 $\frac{9}{20}$ $3, 5, 6, 9, 10, 12, 15, 18, 20$

19 $x = 130$ $2x - 45 = 215$

20 4π **m** $C = 2\pi r = 2\pi \times 2$

B $x = 5$ **(sides** $5, 12, 13$**)** $x^2 + (x + 7)^2 = (x + 8)^2 \Rightarrow x^2 - 2x - 15 = 0$